VIT BHOPAL

CSA2001 - FUNDAMENTALS OF AIML

PROJECT REPORT

CREDITWISE LOAN SYSTEM

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Executive summary

Through this project, we are introducing an automated system for preliminary screening of home and personal loan applications. It is intended to take the place of the conventional manual review procedure, which frequently has biases, inconsistencies, and protracted delays. The **CreditWise Loan System** is an end-to-end Machine Learning-powered software designed to automate and optimize the loan approval process for financial institutions. By leveraging supervised learning classification algorithms, the system evaluates an applicant's demographic, financial, and credit data to predict loan eligibility (Approved or Rejected) in real-time. The goal is to minimize human error, reduce defaults, and accelerate decision-making without sacrificing risk management.

Problem statement

In the current setup, loan officers manually assess factors like income and credit history every day. This leads to several issues such as :

- Qualified applicants getting unfairly rejected.
- Risky cases slipping through, resulting in higher default rates.
- Frustatingly slow turnaround times that hurt customer satisfaction.
- Potential biases that could expose the institution to regulatory risks.

The approach in this project is to device a data-driven tool that flags high-confidence cases early, streamlining the process while promoting fairness.

Dataset description

We worked with a dataset of historical loan records, featuring a binary target variable Loan_Approved (1 for approved, 0 for rejected) and 20 features in total. Here's a breakdown of the key columns:

Column	Description	Type
Applicant_ID	Unique identifier for the applicant	Identifier
Applicant_Income	Monthly income of the primary applicant	Numerical
Coapplicant_Income	Income of any co-applicant	Numerical
Employment_Status	Salaried, Self-Employed or Business	Categorical
Age	Applicant’s age in years	Numerical
Marital_Status	Married or single	Categorical
Dependents	Number of dependents	Numerical
Credit_Score	Score from credit bureau	Numerical

Column	Description	Type
Existing_Loans	Count of active loans	Numerical
DTI_Ratio	Debt-to-Income ratio	Numerical
Savings	Current savings balance	Numerical
Collateral_Value	Value of any collateral	Numerical
Loan_Amount	Amount requested for the loan	Numerical
Loan_Term	Loan duration in months	Numerical
Loan_Purpose	Purpose: Home, Education, Personal or Business	Categorical
Property_Area	Location: Urban, Semi-Urban, or Rural	Categorical
Education_Level	Graduate, Postgraduate or Undergraduate	Categorical
Gender	Male or female	Categorical
Employer_Category	Government, private or self-employed	Categorical
Loan_Approved	The target outcome	Target

Methodology

Data loading and exploration :

We started by using Pandas to load the dataset. From there, we examined its structure (shape and data types), ran basic stats with `df.decribe()` and checked for missing values using `df.isnull().sum()` .

Data Preprocessing :

To handle missing data, we applied SimpleImputer—using the mean for numerical features and the most frequent value for categorical ones. We also dropped the Applicant_ID column to prevent any data leakage.

Exploratory data analysis (EDA) :

We visualised the data to gain insights. Used pie charts to show the balance of approved vs. rejected loans.

Encoding :

For label encoding, we targeted Education_Level and Loan _Approved . Categorical variables like Emplyment_Status, Marital_

Status, Loan_Purpose, Property_Area, Gender and Employer_Category were one-hot encoded (dropping the first category to avoid multicollinearity).

Feature Engineering :

To capture non-linear relationships, we created new features: DTI_Ratio_sq (squared Debt-to-Income ratio) and Credit_Score_sq (squared Credit Score). We removed the original columns afterward to keep things clean and avoid redundancy.

Train - test split and scaling :

With random_state=42 for reproducibility, the data was divided into 80% training and 20% testing sets. After that, we transformed both sets using StandardScaler, which we only fitted to the training data.

Model training and Evaluation

Model	Key Characteristics	Selected ?
Logistic Regression	Simple linear model, quick to train	No

While Accuracy is our baseline, the project prioritizes the balance between two critical risks:

- Precision (Avoiding False Positives) : Crucial because approving a "High Risk" applicant leads to direct financial loss (defaults).
- Recall (Avoiding False Negatives) : Important because rejecting "Low Risk" applicants results in lost revenue and poor customer experience.

Evaluation Metrics

Metric	Formula	Business Significance
Precision	TP / (TP + FP)	Reduces false approvals and default risks
Recall	TP / (TP + FN)	Catches more eligible applicants
F1 Score	2 x (Precision x recall) / (Precision + recall)	Balances precision and recall
Accuracy	(TP + TN) / Total	Gives a general sense of model reliability

Technology Stack

Total/Library	Version	Purpose
Python	3.x	Main programming language
Pandas	Latest	Data manipulation and analysis
NumPy	Latest	Numerical computations
Matplotlib	Latest	Basic plotting and visualization
Scikit-learn	Latest	Machine learning workflows
Jupyter Notebook	Latest	Interactive development environment

Key Findings

- The model identified credit score and DTI ratio as the most influential features. A high credit score signals historical reliability, while a low DTI ratio indicates the borrower has sufficient "breathing room" in their budget to take on new debt.
- By squaring these features , we allowed the model to capture non - linear relationships. For example, the risk doesnt just increase linearly as a credit score drops, it often accelerates exponentially below certain thresholds. This mathematical "stretching" helps the model distinguish between "average" and "prime" borrowers more effectively.
- The dataset contained significantly more Approved cases than Defaulted cases. This imbalance caused the model to be biased toward the majrority class, negatively impacting Recall.
- People who applied with more savings and less debt were much more likely to be approved

Recommendations

Priority	Improvement suggestion	Expected benefit
High	Try ensemble methods like random forest or XGBoost	Boost overall accuracy and robustness
High	Address imbalance with SMOTE or class weights	Better recall without sacrificing precision
Medium	Use GridSearchCV for hyperparameter tuning	Fine-tune models for peak performance
Medium	Apply log transformations to skewed income data	Improve normality and model assumptions
Medium	Incorporate SHAP for model explainability	Help stakeholders understand decisions
Low	Deploy via Streamlit or Flask	Enable real-time, user-friendly access
Low	Integrate ML flow for experiment tracking	Ensure reproducibilty in future work

Conclusion

This project demonstrates how Machine Learning can fundamentally transform the financial landscape by making loan approvals faster, fairer and more accurate. By starting with a Logistic Regression model, we established a transparent and low risk baseline that is easy for regulators and banks to trust. By introducing advanced techniques such as squaring features to find non-linear risk patterns and balancing the dataset to better predict defaults, we have created a system that is credible enough for real-world use. The automated approach moves away from rigid, manual paperwork an toward a data-driven future. It proves that with the right AI tools, we can provide instant financial support to deserving individuals while strictly protecting the lender expensive defaults.